

Software Evolution & Technical Debt

CMP9134: Software Engineering

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Today's Agenda

- 1 Formalizing DevOps & Introduction
- 2 Software Evolution
- 3 Technical Debt
- 4 Code Smells
- 5 Refactoring
- 6 Next Steps

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The Journey So Far

Over the past 11 weeks, we have covered the journey of **building** software:

- **Weeks 1-3:** Agile Planning, Requirements, and UML Models.
- **Weeks 4-6:** System Architecture, Design Patterns, and HCI.
- **Weeks 7-9:** Containerisation (Docker) and Automated Testing.

You have built your Ground Control Station. Your tests are passing. Your containers are running. Before we look at what happens *next*, we need to formally name the methodology you have been using for the past month: **DevOps**.

The Traditional "Wall of Conflict"

Historically, software companies had two entirely separate teams with conflicting goals:

Development (Dev)

Goal: Change & Agility.

"We want to ship new features to users as fast as possible!"

VS

Operations (Ops)

Goal: Stability & Uptime.

"Every time you give us new code, the servers crash. Stop changing things!"

The Result: Developers would write code, throw it over the "Wall of Conflict" to Operations, and wash their hands of it. Deployments took months and frequently failed.

What is DevOps?

Definition

DevOps is the combination of cultural philosophies, practices, and tools that increases an organization's ability to deliver applications at high velocity. Focus: **Collaboration** between Dev and Ops, and **Automation** of the software delivery process.

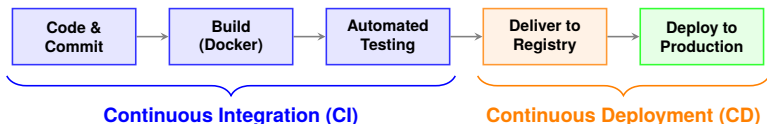
It removes the wall. The engineers who write the code are also responsible for deploying and maintaining the code.

How you already applied this:

- You acted as **Devs** when you wrote your FastAPI/Node.js endpoints.
- You acted as **Ops** when you wrote your `Dockerfile` and `docker-compose.yml` to define the server infrastructure as code!

Formalizing CI / CD

The primary engine of DevOps is the CI/CD Pipeline. You built the first half of this in Week 8 using GitHub Actions.



- **CI (Continuous Integration):** Automatically merging, building, and testing code to ensure the 'main' branch is never broken.
- **CD (Continuous Deployment):** Automatically releasing that tested code directly to the live servers without human intervention.

The Consequence of High Velocity

By using DevOps and CI/CD, companies like Netflix and Amazon can deploy new code to production thousands of times a day.

The New Problem

If you can write, merge, and ship code instantly, the codebase grows and changes at an unprecedented rate. How do you prevent it from turning into a chaotic, unmaintainable mess?

This brings us to the core topic of today's lecture: **Software Evolution and Technical Debt.**

Today's Learning Objectives

By the end of this lecture, you should be able to:

- 1 **Understand** the principles of Software Evolution and Lehman's Laws.
- 2 **Define** Technical Debt and analyze its impact on software lifecycles.
- 3 **Identify** common "Code Smells" that indicate poor design.
- 4 **Apply** Refactoring techniques to improve code quality without altering external behavior.

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The Reality of Software

Software is not a physical bridge. When a bridge is built, it remains static until it decays. Software, however, is a living entity.

Software Evolution

The process of developing, maintaining, and updating software *after* its initial release to accommodate changing user requirements, hardware updates, or business goals.

Why does software evolve?

- The business environment changes (e.g., new robotics regulations).
- Underlying platforms change (e.g., Python 2 is deprecated for Python 3).
- Users discover new ways to use the system that developers never anticipated.

Lehman's Laws of Software Evolution

In the 1970s, Meir Lehman proposed a set of laws regarding software evolution. They remain universally true today.

- 1. The Law of Continuing Change:** A system must continually adapt to its changing environment, or it will become progressively less useful until it dies.
- 2. The Law of Increasing Complexity:** As a system evolves, its complexity increases unless explicit work is done to maintain or reduce it.
- 3. The Law of Declining Quality:** The quality of a system will appear to decline over time unless it is rigorously maintained and adapted to changes in its operational environment.

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What is Technical Debt?

As complexity increases (Lehman's 2nd Law), teams often take shortcuts to deliver features faster. This introduces **Technical Debt**.

The Financial Metaphor (Ward Cunningham)

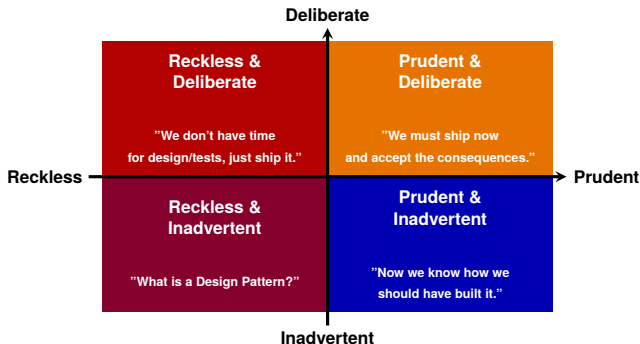
Shipping first-time code is like going into debt. A little debt speeds development so long as it is paid back promptly with a rewrite. The danger occurs when the debt is not repaid. Every minute spent on not-quite-right code counts as **interest** on that debt.

How do we pay the "interest"?

- Slower development times for future features.
- Increased bug rates.
- Developer burnout and frustration.

The Technical Debt Quadrant (Martin Fowler)

Not all technical debt is bad. Sometimes it is a strategic business decision.



Knowledge Check: Classify the Debt

Scenario: Your team is building the Ground Control Station. The deadline is tomorrow. You realize your database queries are unoptimized and might slow down if 1,000 robots connect. You decide to ship the code as-is to meet the deadline, but you immediately create a Jira ticket to rewrite the queries next sprint.

Question: Which quadrant of Technical Debt is this?

- 1 A) Reckless & Deliberate
- 2 B) Prudent & Inadvertent
- 3 C) Prudent & Deliberate
- 4 D) Reckless & Inadvertent

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- 3 C) Prudent & Deliberate
- 4 D) Reckless & Inadvertent

Answer: C (Prudent & Deliberate)

You knew exactly what the consequences were (Deliberate), and you made a calculated business decision while planning for the payback (Prudent).

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Identifying Debt: Code Smells

How do you know if your codebase has technical debt? You look for **Code Smells**.

A code smell is a surface indication that usually corresponds to a deeper problem in the system. It is not necessarily a bug—the code might work perfectly—but it indicates weaknesses in design that increase the risk of future bugs.

Common Code Smells

- **The God Class:** A single class that controls too much, has too many dependencies, and thousands of lines of code. It violates the Single Responsibility Principle.
- **Long Method:** A function that spans multiple screens. It is doing too many things, making it impossible to unit test effectively.
- **Duplicated Code (Copy-Paste Programming):** The same logic exists in three different places. If a bug is found, you have to remember to fix it in all three places.
- **Shotgun Surgery:** Every time you want to make a simple change (e.g., adding a new robot sensor), you have to make tiny edits to 15 different files.
- **Magic Numbers:** Hardcoded numbers scattered through the code (e.g., `if (status == 4)` instead of `if (status == ROBOT_IDLE)`).

Interactive Task: Spot the Smells

Take 60 seconds. Read the code below. How many distinct "Code Smells" can you find?

```
def do_stuff(x):  
    if x == 1:  
        # connect to DB  
        db.execute("UPDATE robots SET status = 'active' WHERE id = 1")  
    elif x == 2:  
        # connect to DB  
        db.execute("UPDATE robots SET status = 'active' WHERE id = 2")  
    # ... repeats 10 more times ...  
    print("Done checking. Now let's calculate battery...")  
    # ... 50 more lines of battery calculation logic ...
```

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    # ... repeats 10 more times ...  
    print("Done checking. Now let's calculate battery...")  
    # ... 50 more lines of battery calculation logic ...
```

The Smells:

- **Magic Numbers:** What do 1 and 2 mean?
- **Duplicated Code:** DB connection and update logic is copy-pasted.
- **Long Method / God Class:** Doing DB updates AND battery calculations in one function called `do_stuff`.

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The Solution: Refactoring

If Technical Debt is the disease, Refactoring is the cure.

Refactoring Definition

A change made to the internal structure of software to make it easier to understand and cheaper to modify **without changing its observable behavior**.

The Golden Rule of Refactoring: You cannot safely refactor code if you do not have an automated test suite (Week 8)! If you change the internal structure without tests, you are just blindly hoping you didn't break anything.

Refactoring Example: Extract Method

The Smell: Long Method with unclear logic.

```
# BEFORE REFACTORING
def process_robot_telemetry(data):
    # Calculate battery percentage
    voltage = data['voltage']
    percentage = (voltage - 10.5) / (12.6 - 10.5) * 100
    if percentage > 100: percentage = 100
    if percentage < 0: percentage = 0

    # Save to database
    db.connect()
    db.execute(f"INSERT INTO telemetry (battery) VALUES ({percentage})")
    db.close()
```

Think-Pair-Share: How do we fix this?

The Bad Code:

```
def process_robot_telemetry(data):  
    voltage = data['voltage']  
    percentage = (voltage - 10.5) / (12.6 - 10.5) * 100  
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Task (2 minutes): Turn to the person next to you. If you had to refactor this using the "Extract Method" technique, what would your new function signatures look like?

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Task (2 minutes): Turn to the person next to you. If you had to refactor this using the "Extract Method" technique, what would your new function signatures look like?

(Let's look at the solution on the next slide...)

Refactoring Example: Extract Method

The Fix: Extract the logic into smaller, testable, named methods.

```
# AFTER REFACTORING
def process_robot_telemetry(data):
    percentage = calculate_battery_percentage(data['voltage'])
    save_telemetry_to_db(percentage)

def calculate_battery_percentage(voltage):
    percentage = (voltage - 10.5) / (12.6 - 10.5) * 100
    return max(0, min(percentage, 100))

def save_telemetry_to_db(percentage):
    db.connect()
    db.execute("INSERT INTO telemetry (battery) VALUES (?)", (percentage,))
    db.close()
```

Behavior is identical. Readability, security (SQL injection fix), and testability are drastically improved.

When should you refactor?

You don't need to schedule a "3-month refactoring sprint". Refactoring should be a continuous, daily habit.

- **The Rule of Three:** The first time you do something, just do it. The second time, wince at the duplication, but do it. The third time you do the same thing, refactor it.
- **When adding a feature:** If the code is too messy to easily add a new feature, refactor the code first to make the addition easy.
- **The Boy Scout Rule:** "Always leave the campground cleaner than you found it." If you open a file to fix a bug and see a poorly named variable, rename it before you commit.

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This Week's Lab: Paying off your Debt

You have been building your Assessment 1 codebase for several weeks now. It is highly likely you have accumulated Technical Debt.

In the workshop, you will:

- **Task 1: Static Analysis:** Install and run a static analysis tool (like SonarLint, ESLint, or Pylint) on your repository to automatically detect Code Smells.
- **Task 2: Identification:** Identify at least 3 major instances of Technical Debt in your own code.
- **Task 3: Refactoring:** Apply formal refactoring techniques to clean the code, ensuring your automated CI tests still pass afterward!

Reading & Resources

- **Recommended Reading:** Sommerville, *Software Engineering 9th Edition*, Chapter 9 (Software Evolution).
- **Refactoring: Improving the Design of Existing Code** by Martin Fowler (The definitive guide to Code Smells and Refactoring techniques).
<https://github.com/pratik0197/books/blob/master/software-dev/Refactoring%20%20Improving%20the%20Design%20of%20Existing%20Code.pdf>
- **Refactoring.Guru:** An excellent, visual website detailing dozens of specific code smells and how to fix them.

Any Questions?

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